

Solution Requirements

Date	10 july 2026
Team ID	XXXXXX
Project Name	EV Data Analytics & Visualization Dashboard
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Allow users (Admin/Analyst) to securely log in and access dashboards based on their roles.</i>	Medium
2	Authorization levels	<i>Admin can manage datasets and dashboards. Analysts can view reports and perform analysis.</i>	Medium
3	External interfaces	<i>Connect MySQL database with Tableau and support EV dataset import (CSV/Excel).</i>	High

4	Transactions processing	<i>Validate, clean, preprocess, and analyze EV charging, battery, and vehicle data before visualization.</i>	High
5	Reporting	<i>Display interactive dashboards for charging patterns, battery performance, EV model comparison, KPIs, and allow report export.</i>	High
6	Business rules, etc.	<i>Calculate charging cost, battery efficiency, driving range, energy consumption, and generate visual insights.</i>	High
7	Compliance to laws or regulations	<i>Ensure secure storage of EV data and role-based dashboard access.</i>	Medium
8	<i>Other</i>	<i>Support interactive filters, search, sorting, and dashboard customization for better analysis.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>Load Tableau dashboards and analytics quickly.</i>	<i>Response time less than 3 seconds.</i>

2	Scalability	<i>Support large EV datasets and multiple users.</i>	Support 500+ concurrent users.
3	Security & Data Privacy	<i>Protect EV datasets using secure communication and authentication.</i>	<i>HTTPS enabled and secure data storage.</i>
4	Reliability & Availability	<i>Ensure the dashboard is always available for users.</i>	99% uptime.
5	Usability & Accessibility	<i>Provide a responsive interface for desktop, tablet, and mobile devices.</i>	<i>Fully responsive UI with clear navigation.</i>
6	Other	<i>Use modular code and well-documented dashboards for future enhancements.</i>	<i>Clean project structure with documented code.</i>